SIH 2026 / PS 26262

# Implementation vs. AICTE Ecosystem

## Deep comparison, critical gaps, and decisions for the next team discussion

**Selected problem statement**

### Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System

| | |
|---|---|
| **Prepared for** | Team architecture and implementation discussion |
| **Reviewed on** | 23 October 2026 |
| **Assessment basis** | Team consolidated engineering specification (22 Oct) + implementation priorities (23 Oct) + publicly available official sources |

**Executive position**

Do not build a second official approval portal. Position the prototype as an integration-ready monitoring, compliance and workflow-coordination layer around AICTE's existing authorised process. The strongest engineering value is rule-source transparency, safe route enforcement, scoped re-scrutiny, approval-to-obligation continuity, and dependable alerts—not basic online forms or a generic dashboard.

Important scope note: this is a design-level gap analysis based on the team's consolidated specification and prioritized hardening guide. The live code repository, authenticated AICTE officer-side screens, and production API contract were not available for a line-by-line implementation audit. "Missing" therefore means not yet evidenced in the reviewed design/proof, unless explicitly described as absent from the public evidence. Scope control: the PS 26262 text supplied for this review explicitly includes lease/document validity but does not list security-deposit/FDR management; FDR appeared in a separate candidate-PS excerpt. Treat FDR-specific features as out of scope unless the final official PS wording confirms them.

01 / DECISION

# Executive assessment

Use this page as the starting point for the discussion. The detailed evidence and action plan follow.

**Overall judgement**

The team's proposed product concept is directionally aligned with PS 26262 and can fit the AICTE ecosystem if it remains a companion layer rather than a parallel source of official truth. The architecture is thoughtful, but its credibility now depends on implementation evidence: backend authorization, rule provenance, document states, alert cancellation, and a credible integration boundary.

## What the current blueprint gets right

- Treats AICTE approval as a regulated, category-dependent workflow rather than a single linear form pipeline.
- Keeps official decisions with the competent human authority; software checks prerequisites and records evidence but does not grant or withdraw approval by itself.
- Models scrutiny, re-scrutiny, conditional EVC work, post-approval obligations, document versions, reminders and a case timeline as connected events.
- Recognises that public AICTE documentation proves some portal capabilities, but does not prove which internal dashboards, escalation rules or production APIs exist.
- Uses a modest modular-monolith design with a relational database, explicit state guards, and a durable alert/outbox pattern rather than adding a large workflow engine or speculative AI.

## The main risks to address

| Risk | Why it matters now | Required response |
|---|---|---|
| R1 — Parallel-system risk | A second portal can create duplicated records, inconsistent statuses and a second manual workload. | Define the product as a companion monitoring layer; identify the authoritative source for every field. |
| R2 — Regulatory route risk | The route depends on category, academic session, conditions and current amendments; wrong branching undermines the entire system. | Source-link each route, pin the applicable rule version, and pause unresolved cases for policy review. |
| R3 — Unproven implementation | A detailed specification is not evidence that backend guards and scheduled jobs work. | Demonstrate repeatable API-level tests and an end-to-end scenario, not UI-only screenshots. |
| R4 — Integration dependency | No verified public contract for all required internal AICTE case events was found in the reviewed source set. | Use a mock/import adapter with clear labels; do not claim live integration until authorised. |

## Priorities for the next iteration

**Hardening order:** P0 backend access control → P1 source-linked route reasoning → P2 document lifecycle → P3 reliable alerts. In parallel, confirm the mock/import adapter contract and distinguish simulated from live data.

02 / EXISTING SYSTEM

# What AICTE already does—and what remains unknown

This distinction is essential to avoid claiming that the team is replacing or inventing capabilities already present in the official ecosystem.

## A. Verified public baseline: AICTE Approval Process Handbook

- AICTE already operates a web-portal-based approval process. The APH describes application evaluation through scrutiny/re-scrutiny and further stages as applicable.
- The handbook states that committee members for scrutiny/re-scrutiny and EVC are selected through web-portal-based automated processes made available by AICTE.
- Deficiencies identified during scrutiny are communicated through the AICTE Web-Portal; re-scrutiny is expressly scoped to deficiencies identified by the scrutiny committee under APH §1.8(g).
- The APH says scrutiny, re-scrutiny and EVC are generally conducted online, with offline handling possible in extraordinary circumstances, and that processing should be recorded for transparency and accountability.

Source basis: AICTE Approval Process Handbook 2024–2027, especially §1.8 and §1.9 [S1]. The handbook's base text must be read together with any currently effective addenda, corrigenda, errata and public notices for the application session in scope.

## B. Verified public baseline: National Single Window System (NSWS)

- NSWS provides an applicant-facing place to discover and apply for participating regulatory approvals and view application updates on the user dashboard.
- NSWS states that after submission, the application is forwarded to the relevant Ministry/Department/State system for internal processing; the final approval decision belongs to that authority, not NSWS.
- This is a useful model for separation of applicant entry/tracking from the authority's internal case-processing system. It does not prove that the specific AICTE internal events needed by PS 26262 are exposed through an API.

Source basis: NSWS FAQ [S2]. AICTE's published FAQ has historically described an NSWS-to-AICTE portal route for specified institution/application journeys [S3]; validate that route for the exact current application category before relying on it.

## C. Do not overstate the evidence

| Publicly established | Not established by public material reviewed | Implication for the project |
|---|---|---|
| Portal-based approval processing, scrutiny/re-scrutiny and EVC mechanisms are documented. | Exact current officer-side dashboards, workload filters, SLA timers, recipient matrices and escalation screens. | Do not assert these tools are absent. Treat them as discovery questions. |
| NSWS documents tracking and hand-off to authority systems for participating approvals. | A public AICTE API contract for every application, stage event, lease/document record and officer assignment needed by this prototype. | No direct live integration claim without an API/data-sharing agreement. |
| The APH defines regulatory requirements and records scrutiny/re-scrutiny/EVC actions. | AICTE internal database schema, deployment topology, identity provider, internal SLA configuration or operational procedures not described publicly. | Keep the design provider-neutral and mark integration assumptions explicitly. |

03 / OUR PROPOSED SOLUTION

# What the team's implementation blueprint actually is

This flow is reconstructed from the team's consolidated engineering specification. It describes the intended prototype, not verified completion of every module.

| Step | Proposed behaviour |
|---|---|
| 1. Intake / case record | Create a case from typed input or seeded fictional data; link institution, category, academic session and reference. |
| 2. Policy registry | Bind the case to a reviewed rule version, source clause, applicability conditions and effective period. |
| 3. Route explanation | Select only the permitted route; display "Why this route?" and fall back to "Policy Review Required" when conditions are unresolved. |
| 4. Scrutiny snapshot | Authorised reviewer records deficiencies and signs/locks the initial checklist snapshot. |
| 5. Scoped re-scrutiny | Allow corrections only for recorded deficiency IDs; reject out-of-scope changes at the backend. |
| 6. Conditional EVC / committee work | Create tasks only when the case's verified route and prerequisites require them; record reports as evidence. |
| 7. Human-authorised decision | A competent-authority demo role records an official-style decision event; normal administrators cannot approve. |
| 8. Obligation hand-off | When a decision creates a follow-up obligation, link it to the case with the exact source, responsible party and due-date origin. |
| 9. Document lifecycle | Track Submitted / Pending / Verified / Rejected / Superseded states, version history, validity and renewal evidence. |
| 10. Alerts, dashboards, audit | Use an event-driven timer/outbox worker to deduplicate, cancel stale reminders and update stage-age dashboards and the chronological event history. |

## Intended technical pattern

| Layer | Current blueprint direction | Important boundary |
|---|---|---|
| UI | React + TypeScript + Vite. | UI reflects server-confirmed state; it is not the security boundary. |
| Backend / data | Choose FastAPI/Python or NestJS/TypeScript; PostgreSQL stores case, policy, event and obligation state. | Choose one backend. Simulated records must never appear as official AICTE data. |
| Workflow | Explicit state machine, rule guards, optimistic versioning and source-pinned routes. | Add a full BPMN engine only if real process complexity justifies it. |
| Alerts / integration | Persisted outbox + scheduled worker; retries, deduplication and current-state checks; mock/import adapter initially. | No stale alerts. Replace mock transport only after an authorised API/data contract is available. |

04 / DEEP COMPARISON

# Requirement-by-requirement comparison (I)

Legend: VERIFIED = explicit in public official sources; UNKNOWN = public sources do not settle current internal coverage; PROPOSED = described in the team blueprint; PROVE = must be demonstrated in the running prototype.

| Capability | Actual ecosystem baseline | Team blueprint | Gap / decision |
|---|---|---|---|
| Application entry and identifiers | AICTE has a web-portal approval process; NSWS has an applicant dashboard and forwards participating approvals to authority systems. [S1, S2] | Create a local case record and unique ID; current MVP permits typed/seeded data. | PROVE + INTEGRATION: keep the authoritative application ID when one exists; do not create a conflicting second application identity. |
| Application-category routing | APH uses category-specific procedures and processing tables; not every case follows the same route. [S1] | Versioned policy registry and route selection by category/session/conditions. | PROVE: at least two source-linked routes; do not collapse all cases into Scrutiny → EVC. |
| Rule source and amendment history | APH is the regulatory baseline; applicable amendments and errata may modify provisions. [S1, S6, S7] | Policy registry with source citation, effective dates, version pinning and approved activation. | HIGH GAP IF NOT LIVE: “Why this route?” must show exact source + session + conditions. Ambiguity must pause the case. |
| Scrutiny / deficiency communication | APH documents scrutiny and deficiency communication through AICTE Web-Portal. [S1 §1.8] | Reviewer checklist, signed deficiency snapshot, corrections and auditable history. | INCREMENTAL: build better provenance and rule-safe follow-up, not a replacement for the existing portal upload process. |
| Re-scrutiny limits | APH §1.8(g) states re-scrutiny verifies only deficiencies pointed out by scrutiny. [S1] | Backend guard allows only flagged deficiency IDs; invalid mutation is logged. | TOP DIFFERENTIATOR TO DEMONSTRATE: show valid D1/D2 correction succeeds while cleared C3 mutation fails through the API. |
| EVC task coordination | APH documents portal-based automated EVC member selection and EVC after applicable processing. [S1 §1.9] | Eligibility queue, task scheduling/status/report and source-guarded EVC creation. | PROVE CONDITIONALITY: no EVC task for a route where the current applicable rule does not require one; don’t recreate committee constitution without authorization. |
| SHC / SAC / appeal stages | APH defines separate hearing/compliance and appellate paths; the route is rule-dependent. [S1] | Conditional committee states are possible in the design. | GAP TO VERIFY: explicitly model SHC and SAC as different concepts; only create the relevant stage when the cited rule and trigger apply. |

04 / DEEP COMPARISON

# Requirement-by-requirement comparison (II)

These rows focus on the explicit cross-cutting requirements in PS 26262: responsibility, compliance, document validity, alerts and operational visibility.

| Capability | Actual ecosystem baseline | Team blueprint | Gap / decision |
|---|---|---|---|
| Officer / Region pendency | The problem statement asks for cross-Region tracking; the exact internal tools available to AICTE officers are not publicly established. | Dashboard for stage/region/officer, age, tasks, overdue and compliance status. | PROVE WITH REAL STORED EVENTS: counters must recalculate from the database, not static demo numbers. |
| Institutional undertakings / ongoing obligations | PS 26262 calls out compliance/undertaking tracking; full current portal coverage is not confirmed publicly. | Approval-linked obligation ledger with source, owner, due date, evidence, review and closure. | CORE VALUE: represent application decision and continuing obligation as linked but separate lifecycles. |
| Lease documents and renewals | Need is explicit in PS; public sources reviewed do not establish a unified cross-case lease-validity workflow. | Document versions and validity dates, renewal status and reminders. | PROVE: expiry date is not equivalent to verified legal validity; keep reviewer status and date provenance. |
| Security deposit / FDR (scope check) | The PS 26262 text supplied for this review does not explicitly list security deposits/FDRs; a separate candidate-PS excerpt shared by the team did. Public documents alone do not establish the current internal FDR interface. | The broader blueprint includes possible deposit/FDR records, maturity dates and replacement/renewal tracking. | SCOPE GATE: do not treat FDR management as a core PS 26262 requirement unless the final official statement explicitly includes it. If not included, remove it from prototype scope, acceptance tests and the pitch. |
| Notifications / escalations | NSWS describes some applicant dashboard updates/notifications; exact AICTE officer-side alert and escalation coverage is unknown. [S2] | Timer/outbox worker, reminders, escalations, delivery status and cancellation. | P3 HARDENING: idempotency key, re-check state at send time, suppress stale messages, record retries/failures. |
| Document verification | Existing portal accepts/uses uploaded evidence within regulated processing. [S1] | Separate Submitted, Under Review, Verified, Rejected and Superseded states; preserve versions. | P2 HARDENING: upload must not equal compliance. Only an authorised reviewer can verify; previous evidence remains auditable. |
| Audit trail / accountability | APH requires recording scrutiny/re-scrutiny/EVC processing for transparency and accountability. [S1] | Actor/time/source-linked event timeline and event/outbox records. | PROVE: successful AND denied actions are logged; separate external system facts from local simulation. |
| Integration with live AICTE / NSWS | API Setu promotes government API interoperability, but the relevant provider must publish/approve the specific API. [S4, S5] | Mock/import adapter in prototype; provider-neutral integration boundary. | HARD DEPENDENCY: no live API contract/credentials verified in reviewed material. Do not scrape the portal or claim production integration. |

05 / INTERPRETATION

# What is genuinely different—and what is not a USP

The team should present the product as a precise integration and reliability improvement, not as the first digital AICTE approval system.

## Baseline capabilities: do not claim these as new

- Online/portal-based application processing, scrutiny/re-scrutiny, deficiency communication and EVC mechanisms are already documented by AICTE. [S1]
- Applicant dashboard tracking and authority-side processing hand-off are established NSWS concepts for participating approvals. [S2]
- Document upload, notifications, stage cards, approval status and generic dashboards are common capabilities. They may be required in the product, but are not credible differentiators on their own.
- The regulatory requirement to limit re-scrutiny to identified deficiencies is not invented by the team. The differentiator is to make that rule explicit, explainable and enforced in the prototype's backend.

## Credible proposed differentiators

| Differentiator | Why it adds value | How to prove it in the demo |
|---|---|---|
| 1. Source-linked route reasoning | A regulator can see which session-specific clause and conditions selected a route; unresolved cases are not silently guessed. | Open two differently routed fictional cases side by side; show clause, policy revision, conditions and "Policy Review Required" for a boundary case. |
| 2. Enforced differential re-scrutiny | Makes the existing scope restriction visible and robust at the API layer. | API request changes flagged D1/D2 successfully; attempt to change accepted C3 returns a safe 403/409 and creates an audit event. |
| 3. Approval-to-obligation continuity | The end of one application workflow creates a traceable follow-up lifecycle where the rule requires it. | A human-authorised decision creates one linked obligation, evidence-review task and a due-date timer with source provenance. |
| 4. Exception-aware alerts | Prevents stale or duplicate alerts when evidence is verified, a due date changes, or an obligation is cancelled. | Trigger, retry, then complete/renew the obligation; demonstrate a single logical alert and suppression of the obsolete reminder. |
| 5. Joined operational view | Lets authorised users see case stage, responsible owner, due items and event history without presenting local data as an official decision. | Move cases between states and show dashboard counts/age plus the same events in audit history. |

### Novelty wording for the pitch

"We are not replacing AICTE's approval portal. We propose a policy-aware coordination layer that makes case routing explainable, enforces the scope of re-scrutiny in the backend, links authorised decisions to ongoing compliance obligations, and keeps reminders consistent with the latest evidence and due dates."

06 / MUST-FIX GAPS

# P0–P3: implementation-hardening priorities

These items already appear in the team's Implementation Priorities guide. Treat them as correctness requirements, not optional polish.

## P0 — Backend access-control and data isolation

- Concern: a role-based UI is insufficient if the backend accepts another institution's or Region's identifier from the browser.
- Implement/check: authenticate the actor; derive institution/Region scope from the session; enforce permission on every case read/update, evidence upload and document download; default-deny; separate competent-authority decision rights from a normal admin.
- Acceptance proof: Institution A can read/edit/download only its own case and files. Changing the case ID to Institution B's ID fails without data leakage; the failure is safely logged.

## P1 — Explainable, version-pinned regulatory routing

- Concern: a hard-coded linear workflow will be wrong for conditional application categories or amended rules.
- Implement/check: each route stores the application category, academic session, relevant conditions, source URL/clause, policy revision, effective period, approver/reviewer and selected stage path. Separate source-verified rules from simulated demo timers/policies.
- Acceptance proof: two source-backed cases choose different paths. A missing, ambiguous or boundary condition returns "POLICY REVIEW REQUIRED"; no speculative stage is generated.

## P2 — Document lifecycle and independent verification

- Concern: uploaded evidence can be mistaken for accepted compliance, and renewal can overwrite prior evidence.
- Implement/check: explicit document states; link each version to its obligation and document type; capture upload time, claimed validity dates, verification time and reviewer; keep prior versions; restrict files by role and institution.
- Acceptance proof: upload remains Submitted/Pending; only an authorised reviewer can verify/reject; replacement evidence leaves the earlier version and its history viewable to permitted roles.

## P3 — Retry-safe and non-stale alerts

- Concern: repeated worker retries can duplicate reminders, while a message may still be sent after renewal, correction or completion.
- Implement/check: stable deduplication key tied to obligation and timer generation; validate the latest status immediately before send; cancel/supersede obsolete timers; persist delivery attempts and failures.
- Acceptance proof: trigger one alert event, retry it multiple times, then verify/renew the obligation before delivery; only one valid message is delivered and the stale message is suppressed.

**Recommended order**

P0 first, then P1, P2 and P3. Do not trade these away for extra charts, AI features or a more elaborate workflow engine. The current prototype becomes much more credible when these guards work repeatably.

07 / MUST-CLARIFY GAPS

# Additional issues beyond P0–P3

These issues are either incompletely specified in the design or require explicit team confirmation before calling the system ecosystem-ready.

| Gap / question | Why it is important | Recommended change |
|---|---|---|
| P4 — Integration contract and source of truth | No confirmed AICTE production API/data contract is available in the reviewed bundle. NSWS/API Setu are patterns and possible discovery points, not proof that the needed APIs exist. | Write an interface contract for the prototype adapter: event type, source ID, case ID, source timestamp, received timestamp, schema version, signature/auth context and idempotency key. Keep mock data visibly tagged. |
| P5 — Regulatory versioning and change control | The 2024–27 APH is supplemented by session-specific addenda/corrigenda, including an official 2026–27 addendum. Routes must be verified against the applicable version; every other relevant notice should also be checked before rules are frozen. | Create a reviewed policy source register. Pin active cases to a rule version. Activate new versions through review/approval; never silently rewrite existing case history. Keep uncertain clauses on hold. |
| P6 — Stage semantics and authority | SHC, SAC and EVC represent different activities. A general “committee stage” can wrongly conflate compliance hearing with an appeal or imply unauthorized decision powers. | Model stage type, trigger, authority, prerequisites and outputs separately. Final/penalty/withdrawal decisions remain human-authorised, with evidence and actor rights checked server-side. |
| P7 — Time semantics | Public academic deadlines, event-relative durations, internal operational SLAs and document validity are different clocks. | Represent fixed date, trigger-relative interval, validity window, recurring obligation and authorised extension/cancellation separately. Every alert must show how its due date was calculated. |
| P8 — Dashboard truthfulness | Static numbers or UI-local updates can make an attractive dashboard that contradicts case history. | Compute counts and ageing from persisted records/events; test dashboard totals against a database query after staged transitions. |
| P9 — Lease lifecycle + FDR scope check | Lease/document validity is explicit in the supplied PS. The supplied PS text does not explicitly mention FDR/deposit tracking. If FDR is separately confirmed in the final selected PS, its instrument, maturity, bank-confirmation and acceptance states would need a distinct model. | Keep lease expiry, renewal submission and verification in scope. Keep deposit/FDR functionality out of the core build unless the final official PS explicitly includes it; if confirmed, model uploaded/self-declared facts separately from verified financial status and never claim bank verification without an authorized mechanism. |
| P10 — Production readiness | Prototype security, backups, retention, malware scanning, accessibility and operational monitoring do not automatically equal government deployment readiness. | Document production assumptions and a later hardening path. Use synthetic data now; keep files private; add audit, backup/restore and keyboard/accessibility tests suitable for the demo. |

08 / FIT WITH AICTE

# Target ecosystem fit: companion layer, not parallel authority

The product's most important architectural decision is the source-of-truth boundary.

## Recommended boundary

| Information / action | Authoritative owner | What our layer should do |
|---|---|---|
| Official application identity and submitted status | Existing authorised AICTE/NSWS-linked application system, according to the actual journey. | Import/associate the existing reference; preserve source, time and synchronization status. |
| Official scrutiny/EVC findings and reports | Authorised committee/reviewer and the official system of record. | Mirror permitted metadata or attach authorised reference; never invent or silently overwrite official findings. |
| Official approval / refusal / penalty / withdrawal | Competent authority under applicable procedures. | Record a sourced/authorised decision event; do not auto-decide. |
| Compliance obligation register, due-date computation and monitoring views | Proposed coordination layer, based on verified rule triggers and authorised case events. | Add cross-case visibility, explainable timers, owners and evidence-review tracking. |
| Internal notification delivery and monitor audit | Proposed coordination layer for its own alerts/actions. | Record generated/sent/suppressed status and why; identify simulated versus live source events. |
| Policy source version and rule explanation | Approved regulatory source register, subject to owner review. | Display the source and review state; no silent route changes or invented official SLAs. |

## Integration approach

- Now (SIH prototype): use a mock/import adapter and fictional institutions; clearly label source events as SIMULATED. Demonstrate the business rules independently from the transport layer.
- Future (if authorised): implement a provider-approved API, event feed, or controlled import. Store source identifiers, payload/version metadata, received times and idempotency keys; validate the schema and permissions.
- Conflict handling: if an external status and a local monitoring projection disagree, preserve both observations and flag reconciliation. Do not allow the local monitor to overwrite the authoritative status.
- Security: do not rely on scraping, browser automation or unofficial database access as the integration plan. API Setu's official documentation describes discovery/subscription/provider approval; it does not guarantee that an AICTE endpoint is published. [S4, S5]

### Integration is a dependency, not a demo claim

A mock adapter that is designed to be replaced is acceptable for an SIH prototype. Calling the prototype "integrated with AICTE" is not acceptable unless a real, authorised source connection has been verified and demonstrated. Use accurate labels such as "integration-ready prototype" or "simulated AICTE workflow data".

09 / VERIFICATION PLAN

# Acceptance tests that should decide whether a feature is done

Mark “Done” only after each test passes against the running backend on a clean seeded database. A screen looking correct is not sufficient.

| ID | Test | Stimulus | Pass criterion |
|---|---|---|---|
| T01 | Cross-institution read | Institution A requests Institution B’s case. | Denied server-side; no data leakage; denial logged. |
| T02 | Cross-institution document access | Institution A requests B’s evidence file. | Download denied; file remains private; denial logged. |
| T03 | Different route choices | Submit two fictional cases with different source-verified conditions. | Each shows the correct route, source and reason. |
| T04 | Ambiguous policy | Submit a case with unresolved/edge conditions. | “Policy Review Required”; no guessed stage. |
| T05 | Re-scrutiny guard | Initial scrutiny marks D1/D2 deficient and C3 accepted; edit D1/D2 then attempt C3. | D1/D2 correction accepted; C3 rejected by API and audited. |
| T06 | Conditional EVC | Advance a route that does not require EVC. | No EVC task created; applicable route remains valid. |
| T07 | Role boundary | Ordinary admin attempts to record final approval. | Denied; only designated demo decision role succeeds. |
| T08 | Document upload/verify | Institution uploads an obligation document. | Status remains Submitted/Pending until authorised review. |
| T09 | Version retention | Upload replacement evidence. | Old version and dates/reviewer history remain traceable. |
| T10 | Duplicate alert | Retry same event/job three times. | One logical notification delivered per dedupe key. |
| T11 | Stale alert suppression | Queue reminder, then verify/renew or change the due date. | Obsolete reminder cancelled/suppressed before send. |
| T12 | Deadline provenance | Open a generated alert. | Shows obligation, source or simulated rule label, due-date origin, owner and next action. |
| T13 | Concurrent state update | Two updates use the same expected case version. | One succeeds; stale competing update is rejected with consistent audit/outbox. |
| T14 | Dashboard consistency | Move seeded cases across stages. | Counts/ageing equal persisted records and event history. |
| T15 | Source change | Introduce a new policy revision. | Active cases remain pinned unless an authorised migration is recorded. |
| T16 | Integration honesty | Switch source from simulated to adapter test fixture. | Every record shows source type; no mock record appears live/official. |

Recommended dashboard metrics for the demo: route-selection pass rate on a fixed labelled fixture set; invalid-transition rejection count; audit coverage; duplicate/stale-alert test outcomes; deadline calculation correctness against expected fixture values; and exact agreement between dashboard counts and database queries. Do not invent real-world time savings or predictive accuracy without a measured baseline.

10 / TEAM ACTION PLAN

# Suggested meeting agenda and next steps

Use this section to close the discussion with concrete decisions instead of a general list of enhancements.

| Order | Decision / deliverable | Owner to assign | Exit condition |
|---|---|---|---|
| 1 | Confirm the exact existing AICTE/NSWS applicant journey for the target application categories. | Research / product | Category-by-category AS-IS map with evidence links; unknowns marked explicitly. |
| 2 | Freeze the minimum two or three source-verified route fixtures. | Policy research + backend | Each fixture has category/session/conditions/source version and expected stage list. |
| 3 | Complete P0 backend authorization tests. | Backend + security | Cross-institution and cross-role checks pass for reads, edits, uploads and downloads. |
| 4 | Complete P1 route explanation and unresolved-case fallback. | Policy + backend + UI | Backend-selected route and UI explanation always agree. |
| 5 | Complete P2 document state/version handling. | Compliance/document workstream | Upload, review, reject, verify and replace are distinct states with auditable history. |
| 6 | Complete P3 timer dedupe and stale-alert suppression. | Alert/integration workstream | Retry and cancellation scenarios pass reproducibly. |
| 7 | Add mock adapter contract and source labels. | Integration workstream | One simulated case and one imported fixture use the same business logic but remain distinguishable. |
| 8 | Run end-to-end regression and prepare demo narrative. | Integration / QA / pitch | Clean seeded demo reproduces the full lifecycle without manual database edits. |

## Questions the team should explicitly resolve

- Which exact application categories and APH clauses can we confidently demonstrate with current official text? Which need “Policy Review Required”?
- Which status fields are external authoritative data, which are local operational projections, and which are clearly simulated?
- Do we have any verified AICTE endpoint, sandbox or data contract? If not, what exact mock interface will represent it?
- Who is permitted to create/verify obligations, reschedule reminders, edit rule drafts, approve rule versions and record final decisions in the demo?
- Does the final official PS 26262 explicitly include security deposits/FDRs, or did that requirement come from a different candidate PS? Keep lease-document validity in scope; decide FDR fields and workflows only after the official scope is confirmed.
- Can we demonstrate all P0–P3 tests on the real backend before adding the optional Workflow Integrity Checker or any AI/predictive feature?

**Scope freeze:** prove two source-backed routes → scrutiny → scoped re-scrutiny → conditional EVC → authorised demo decision → obligation/document review → one explainable alert → dashboard/audit. Defer predictive AI and complex engines until P0–P3 pass.

11 / GUARDRAILS

# Claims we can make—and claims we must avoid

This section is intended to protect the team from overclaiming during the report, presentation and judge Q&A.

| Safe, supportable statement | Avoid claiming |
|---|---|
| “AICTE’s public handbook documents portal-based approval processing, scrutiny/re-scrutiny and EVC procedures.” | “AICTE has no digital approval workflow.” |
| “The prototype proposes improved source-linked routing, enforced re-scrutiny scope, obligation continuity and dependable reminders.” | “These are completely new features that AICTE does not have.” |
| “The prototype uses clearly labelled fictional cases and simulated integration events until an authorised interface is confirmed.” | “Our platform is live-integrated with AICTE/NSWS” without demonstrated authorised access. |
| “The software blocks invalid transitions and prepares information for human review.” | “The AI/system grants approval, imposes penalties, withdraws approvals, or certifies legal validity automatically.” |
| “The event history is designed to be traceable and tamper-resistant within the prototype.” | “The event log/hash is court-ready legal evidence” without a legal and operational basis. |
| “We will measure correctness on deterministic test fixtures.” | “The system reduces AICTE processing time by X%” without an observed baseline and measured trial. |

## Short glossary for the discussion

| Term | Meaning in this document |
|---|---|
| APH | AICTE Approval Process Handbook; the policy baseline, read with applicable updates. |
| EVC | Expert Visit Committee; a procedural stage when the applicable route requires it. |
| SHC | Standing Hearing Committee; a conditional hearing/compliance-related route, not a synonym for SAC. |
| SAC | Standing Appellate Committee; associated with the appellate route, not a generic compliance review. |
| Policy version pinning | A case keeps the reviewed rule revision that was used to select its route unless an authorised migration is recorded. |
| Outbox | A durable database record of an event/notification to be published; enables safe retries and avoids losing alerts when a process fails. |
| Mock adapter | A test implementation that imitates a source-system interface using fictional data; not a live integration. |
| Source of truth | The system/authority whose record is authoritative for a particular fact or decision. |

12 / EVIDENCE REGISTER

# Official references and review basis

Public references checked for this document on 23 October 2026. Open the linked sources before updating route logic.

**[S1] AICTE, Approval Process Handbook 2024–2027.** https://www.aicte.gov.in/sites/default/files/APH%20Final.pdf
Primary official source for the documented approval process. Particularly relevant: §1.8 (scrutiny/re-scrutiny), §1.9 (EVC), application categories/tables, and later provisions on committee/compliance processes. Read with currently effective addenda and errata.

**[S2] National Single Window System (NSWS), Frequently Asked Questions.** https://www.nsws.gov.in/faqs
Official FAQ describes applicant dashboard/status tracking, forwarding submitted applications to the relevant authority system, and confirms the final decision belongs to the competent Ministry/Department/State—not NSWS.

**[S3] AICTE, published FAQ (historical entry-point reference).** https://aicte.gov.in/sites/default/files/FAQ%20Updated.pdf
Useful to investigate the NSWS-to-AICTE portal journey for the specified application categories. Treat this as historical/current-scope-dependent evidence, not proof of every 2026–27 internal screen.

**[S4] API Setu, official overview and API discovery model.**
https://docs.apisetu.gov.in/document-central/explore-apisetu/Overview.html
Describes API publishing/discovery, consumer registration/subscription and provider approval. This is an interoperability pattern, not evidence that an AICTE workflow API is available to the team.

**[S5] API Setu / MeitY, Government of India Open API Policy.**
https://docs.apisetu.gov.in/document-central/api-policy/Implementation%20Mechanism.html
Reference for government API publishing and interoperability. API access still depends on the publishing authority, specific contract and required permissions.

**[S6] AICTE, public notice index referenced by team research.** https://www.aicte.gov.in/node/3342
Use to re-check the latest revised public notices/addenda relevant to AY 2026–27. The team's consolidated research notes that at least one linked notice body was not fully revalidated; keep dependent rules under review until the original text is checked.

[S7] AICTE, Addendum/Corrigendum 2026–27 of APH 2024–27.
https://aicte.gov.in/downloads/aph/Final%20Addendum%202026-27_Updated.pdf
Official session-specific amendments to the APH. Check the exact changed clause and effective scope before implementing route or deadline rules.

## Internal team documents used as the implementation baseline

- SIH2026_PS26262_FINAL_Consolidated_Master_Report.pdf — research synthesis and engineering specification dated 22 October 2026. It describes the target case lifecycle, proposed architecture, MVP, API examples, tests and integration limitations.
- PS26262_Implementation_Priorities.docx / .pdf — P0–P3 hardening plan dated 23 October 2026: backend access control, explainable rule routing, document lifecycle integrity and reliable alerts.

### Final decision for the team

Build the smallest secure, source-grounded workflow that complements AICTE's existing process. First prove access controls, rule correctness, document lifecycle and alert reliability; in parallel, preserve a mockable integration boundary. Do not add complexity until those properties work and the team can explain exactly what is existing, what is proposed, what is simulated and what requires official access.